Venkata Lakshmi Parimala Pasupuleti

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SUMMARY

Software engineer with 6+ years building production AI automations, LLM agent loops, and backend systems for regulated environments. Embed with business teams, map workflows end-to-end, and build agentic tools that scale their expertise -- the same pattern I used at Entrata to automate housing compliance and HUD data workflows across a 100K+ MAU platform. Built LeaseI, an 8-agent LangGraph system with engineered loop architecture (act, verify against goal, stop or escalate to human) for housing compliance operations. Document extraction, structured outputs, LLM-as-judge evaluation, and security-first design are core to how I build. Currently deploying Claude into financial workflows at Intuit. Daily Claude Code user.

CORE COMPETENCIES

LLM Agent Loop Engineering | Python / FastAPI | TypeScript / Node.js | LangGraph (Multi-Agent) | Document Extraction & OCR | Structured Outputs | LLM-as-Judge Evaluation | Human-in-the-Loop (HITL) | Workflow Automation | REST API Integrations | SQL / PostgreSQL | AWS (Lambda, S3, RDS, SQS) | Audit Logging & Least-Privilege | PHI / Regulated Data Security | CI/CD (GitHub Actions) | Docker | Business Process Mapping | Runbook Documentation

EXPERIENCE

Intuit

Jun 2026 -- Present

Software Engineer -- Virginia, USA

- Build **LLM automations for financial workflows** -- structured-output schemas, prompt design, and retrieval-grounded Claude responses for financial data summaries and natural-language queries.
- Implement **guardrails and audit controls** around LLM calls: input validation, sensitive-data handling, and fallbacks when model output fails checks.
- Build **LLM-as-judge eval harnesses** with golden datasets in CI, catching hallucinations and regressions before release; maintain runbooks for production LLM services.
- Deploy Python/FastAPI services on AWS via Docker and GitHub Actions CI/CD; instrument with Prometheus and OpenTelemetry.

Entrata -- Affordable Housing & Property Management SaaS

Oct 2020 -- Jul 2024

Senior Software Engineer -- Multi-tenant platform, 100K+ monthly active users

- **Embedded with business and compliance teams** to understand housing workflows end-to-end -- lease management, income eligibility, HUD imports, rent calculations -- and built the automation and tooling that scaled those workflows across the platform.
- Owned the annual **HUD income-limit automation** for 6 years: mapped the federal data transformation end-to-end, wrote build-ready specs, built the pipeline (~180,000 domain records), and maintained runbooks for the annual release cycle. Zero compliance violations.
- Built **document automation pipelines**: 56 state-specific housing certification documents generated via automated merge-field workflows; OCR-adjacent document ingestion for resident records.
- Built **RabbitMQ async automation** for gross-rent updates across thousands of tenant records with idempotency guarantees, retry logic, and human escalation for exceptions.
- Implemented **least-privilege access controls and PHI encryption** (Libsodium) covering 100% of resident data in compliance with federal housing standards; led AWS S3 migration with security validation.
- Wrote Python/Bash automation cutting manual operational work by 80%; optimized 200+ SQL queries (30% gain); reviewed 300+ PRs and mentored junior engineers.

Entrata India

Sep 2018 -- Oct 2020

Junior Software Engineer -- Pune, India

- Built property management backend modules (Python, Java, PostgreSQL) and large-scale data reconciliation pipelines processing 1.6M HUD/Tax Credit records with validation and rollback.

PROJECTS

Leasel -- LLM Agent Loop for Housing Compliance

Python · LangGraph · FastAPI · Loop Engineering · HITL

8-agent LangGraph system with engineered loop architecture for housing compliance: each agent has a clear goal, tools, verification against expected results, defined stop conditions, and a human checkpoint wherever a compliance or financial decision is required. Document ingestion (OCR), section-aware chunking, hybrid retrieval, and answers cited to source clause. Audit trail on all agent actions. LLM-as-judge CI: 93.6% precision, 87.9% F1 across 1,240 test cases.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe -- LLM Evaluation & Governance Toolkit

Python · FastAPI · LLM-as-Judge · CI/CD · Audit Logging

Production evaluation framework: 15-endpoint FastAPI backend, LLM-as-judge scoring across five metrics (accuracy, hallucination, retrieval, completeness, compliance), and CI regression against golden datasets. Built to measure every model or prompt change before it ships -- the same pattern Ascent's AI governance gate requires.

github.com/venkatalakshmiparimala/AgentProbe

AI on FHIR -- Clinical NLP Automation

Python · Flask · REST APIs · RBAC · Least-Privilege · Docker

Automated clinical question-to-FHIR-API workflow using LLMs and BERT. Fine-grained role-based access control for HIPAA-style data isolation; secrets management and least-privilege design throughout. Reduced manual clinical lookup time by ~60%.

github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 | GPA 3.87

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017